

A monotonicity preserving conservative sharp interface flow solver for high density ratio two-phase flows



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ABSTRACT

This paper presents a novel approach for solving the conservative form of the incompressible two-phase Navier–Stokes equations. In order to overcome the numerical instability induced by the potentially large density ratio encountered across the interface, the proposed method includes a Volume-of-Fluid type integration of the convective momentum transport, a monotonicity preserving momentum rescaling, and a consistent and conservative Ghost Fluid projection that includes surface tension effects. The numerical dissipation inherent in the Volume-of-Fluid treatment of the convective transport is localized in the interface vicinity, enabling the use of a kinetic energy conserving discretization away from the singularity. Two- and three-dimensional tests are presented, and the solutions shown to remain accurate at arbitrary density ratios. The proposed method is then successfully used to perform the detailed simulation of a round water jet emerging in quiescent air, therefore suggesting the applicability of the proposed algorithm to the computation of realistic turbulent atomization.

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1. Introduction

Since the seminal work of Harlow and Welch [1], the development of numerical techniques for Direct Numerical Computation of realistic free surface flows has been an active area of research. As of now, in spite of the improved predicting capabilities of the algorithms proposed over the years, no standard approach has emerged. Indeed, the numerical representation of two immiscible fluids with different thermodynamic properties is a challenging task, especially in the multiscale flows typically encountered in, for instance, industrial applications.

In spite of steadily increasing computer power, computational cost still remains a limiting factor. An efficient use of computational resources is therefore critical to reach flow conditions encountered in energy conversion devices for instance. Efficient use of a given mesh suggests sharp interface methods, which consist of resolving the interface front over a single cell, as a relevant approach. Such treatment of the interface singularity is obviously prone to stability issues, for it may involve density variations of two or more orders of magnitude. While the use of low pass filters [2] and/or artificial viscosity [3] is often required, in the case of under-resolved computations or de-aliasing, for example, excessive use of such techniques in Direct Numerical Simulations can be detrimental to the quality and dependability of the results.

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This work proposes an algorithm for the solution of the conservative form of the two-phase incompressible Navier–Stokes equations. It relies on three key components: a geometric integration of the convective momentum transport, a monotonicity preserving momentum rescaling, and a consistent Ghost Fluid projection that includes surface tension effects. The resulting geometric discretization is shown to preserve monotonicity for the density, momentum and velocity fields while simultaneously maintaining the second order interface representation required to accurately capture capillary effects. In addition, a discretely equivalent finite volume formulation is provided to characterize the small, third order conservation error.

The algorithm is tested with a variety of cases and shown to accurately represent convection dominant flows as well as viscous and capillary effects. The localized numerical dissipation associated with the proposed discretization is shown to be a first order effect. The primary atomization computation of a water jet emerging in a quiescent air environment is also presented. The proposed method, combined with an efficient pressure solver [4], is able stably handle the resulting large density ratio (800) of this challenging configuration, and the computational time is shown to remain reasonable (of the order of 100,000 CPU hours for a 400M point structured mesh).

2. Governing equations

The following section focuses on the set of equations solved in the one-fluid formulation adopted throughout this work. The presence of two immiscible fluids gives rise to singularities that must be accounted for, and a good understanding of the governing equations is required to ensure consistent discretization.

The incompressible Navier–Stokes equations are first written in strong form, and a brief description of how they are typically solved in a one-fluid formulation is presented. Directly updating the physical (as opposed to conserved) variables using single phase flow discretizations, however, is known to be prone to numerical momentum transfer between phases, potentially leading to unphysical flow features at moderate and high density ratios [5]. Numerical procedures such as TVD Runge–Kutta time integration [6] and upwind discretization of the convective transport [7] have been used to extend the range of application of these non-conservative formulations. Alternatively, discretizations based on the conservative form of the Navier–Stokes equations have been proposed and shown [8,9] to lead to accurate results for arbitrary density ratios. This therefore motivates the derivation of the integral form of the Navier–Stokes equations for the conserved variables, which is presented in a second part. The resulting integral form of the equations is at the core of the current method, and the link with the proposed discretization is introduced in a third and last part, in which a new strategy to preserve monotonicity for density, momentum, and velocity at the cost of a localized introduction of numerical dissipation and a small conservation error is devised.

2.1. Strong form

In the absence of mass transfer between phases, and if both fluids are viscous, the velocity jump across the interface is null. While the thermodynamic properties remain discontinuous, the continuity of the velocity field greatly simplifies the implementation of the momentum equation if the strong non-conservative form of the Navier–Stokes equations is used. This is emphasized in the following section, which introduces the strong forms of the marker and Navier–Stokes equations, and how they are typically solved in the incompressible limit.

An accurate prediction of the interface location is the cornerstone of predictive two-phase flow computations. In front capturing methods, the interface is transported by means of a mapping to an iso-contour of a marker function f . The evolution of f is then governed by

$$\frac{\partial f}{\partial t} + \mathbf{u} \cdot \nabla f = 0. \quad (1)$$

The choice of the numerical scheme used to discretize Eq. (1) depends on the nature of f , which may be chosen to be a smooth (Level Set) or heaviside (Volume-of-Fluid) function for example. In the proposed work, we adopt a Volume-of-Fluid (VoF) viewpoint, in which the marker function f is the indicator function

$$f(\mathbf{x}, t) = \begin{cases} 0 & \text{if } \mathbf{x} \text{ is in the gas phase at time } t, \\ 1 & \text{if } \mathbf{x} \text{ is in the liquid phase at time } t. \end{cases} \quad (2)$$

In VoF methods, in order to limit the numerical diffusion of the interface front, the solution of Eq. (1) follows from a two-step procedure [10]. First, given a field of cell-averaged indicator function values (volume fraction F), a discontinuous reconstruction of the interface is computed. From this representation, the partial differential equation (1) is solved by evaluating the volume fraction fluxes in a geometric (also referred to as sharp) fashion.

In the incompressible limit of the Navier–Stokes equations, the flow field in both gas and liquid phases is governed by

$$\begin{cases} \nabla \cdot \mathbf{u} = 0, \\ \rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{b}, \end{cases} \quad (3)$$

where the viscous stress tensor is denoted $\boldsymbol{\tau}$ and the body forces by $\mathbf{b}$. At the interface, in the iso-thermal limit, gas and liquid flow fields are connected by the Rankine–Hugoniot jump conditions

$$\begin{cases} \llbracket \rho(\mathbf{u} - \mathbf{w}) \cdot \mathbf{n}_I \rrbracket = 0, \\ \llbracket p\mathbf{n}_I - \boldsymbol{\tau} \cdot \mathbf{n}_I + \rho\mathbf{u}(\mathbf{u} - \mathbf{w}) \cdot \mathbf{n}_I \rrbracket = 2\sigma\kappa\mathbf{n}_I, \end{cases} \quad (4)$$

where the jump operator is defined as $\llbracket \varphi \rrbracket = \varphi^L - \varphi^G$. The subscript I denotes the interface, the superscripts L and G the liquid and gas phase respectively, and σ the thermodynamic tension. κ is the arithmetic average of the interface principal curvatures. The interface velocity $\mathbf{w}$ is connected to the evaporation rate

$$\dot{\omega} = \rho^L(\mathbf{u}^L - \mathbf{w}) \cdot \mathbf{n}_I = \rho^G(\mathbf{u}^G - \mathbf{w}) \cdot \mathbf{n}_I, \quad (5)$$

where the normal vector $\mathbf{n}_I$ is directed towards the gas phase.

In the one-fluid formulation used in front capturing methods, the two phases are treated as a single fluid with variable thermodynamic properties that change abruptly across the interface, and the flow solution procedure typically follows a predictor–corrector approach [11].

Such an approach consists in two steps: first, the velocity field is predicted by advancing the convective term along with the other terms in the right-hand side of Eq. (3). In the absence of transfer across an interface separating two viscous fluids, the velocity field is continuous. Most one-fluid formulations use this information to approximate the convective transport term in Eq. (3) regardless of the interface location. The remaining terms in the predictor step, such as surface tension and viscous effects, are handled using dedicated two-phase techniques [12–14]. Such specific treatments are also presented in detail in Section 4.

The second step consists in correcting the predicted velocity in order to satisfy the continuity equation, which for incompressible flows reduces to the divergence-free condition Eq. (3). For two-phase flows, this results in inverting a variable coefficient Poisson system:

$$\nabla \cdot \left(\frac{1}{\rho} \nabla p \right) = \frac{\nabla \cdot \mathbf{u}^*}{\Delta t}. \quad (6)$$

The reader is referred to [15,16] for details. In the present study, this strategy is retained, but slightly modified to make the projection step momentum conservative for staggered grid arrangements.

In previous work [17,5], centered non-conservative formulations have been shown to be prone to numerical instabilities at moderate and high density ratios, hence requiring the use of TVD time integration or upwinding for example to lead accurate solutions in such conditions [18]. Alternatively, discretizations mimicking the integral form of the Navier–Stokes equations, while still projecting the velocity according to Eq. (6), were found to be very robust, even at extreme density ratios [19,17,20,5]. This formulation will be presented in the next section.

2.2. Integral form

Starting from the marker function transport equation, this section focuses on the integral form of the governing equations. For a material control volume $\Omega(t)$ (defined as a volume whose boundaries move with the fluid velocity $\mathbf{u}$), the integral form of the marker transport equation writes

$$\frac{d}{dt} \int_{\Omega(t)} f(\mathbf{x}, t) d\mathcal{V} = - \int_{\Sigma(t)} \frac{\dot{\omega}}{\rho^L} dS + \int_{\Omega(t)} f(\nabla \cdot \mathbf{u}) d\mathcal{V}, \quad (7)$$

where $\Sigma(t)$ denotes the interface surface present in $\Omega(t)$. The terms on the right-hand side are included for the sake of completeness but are assumed to be both zero since no transfer occurs across the interface and the fluids are assumed to be incompressible.

Writing a similar equation for $(1 - f)$ and adding it to Eq. (7) leads to the equation governing the evolution of the volume of a material control volume $\Omega(t)$, denoted $|\Omega(t)|$,

$$\frac{d}{dt} |\Omega(t)| = - \int_{\Sigma(t)} \frac{\dot{\omega}}{\rho_l} dS + \int_{\Omega(t)} \nabla \cdot \mathbf{u} d\mathcal{V}. \quad (8)$$

Eq. (8) simply states that, for an incompressible flow and in the absence of mass transfer across the interface, volume is conserved. This is trivial in an analytical sense, but Eq. (8) will be useful to later interpret the terms arising in the proposed discretization of the convective transport in the marker, mass, and momentum equations.

While the jump conditions (Eq. (4)) still hold, the density and momentum equations may be rewritten in a similar fashion leading to the integral one-fluid formulation

$$\begin{cases} \frac{d}{dt} \int_{\Omega(t)} \rho d\mathcal{V} = 0, \\ \frac{d}{dt} \int_{\Omega(t)} \rho \mathbf{u} d\mathcal{V} = - \int_{\partial\Omega(t)} p \mathbf{n} dS + \int_{\partial\Omega(t)} \boldsymbol{\tau} \cdot \mathbf{n} dS + \int_{\Omega(t)} \rho \mathbf{b} d\mathcal{V}. \end{cases} \quad (9)$$

The notation $\partial\Omega(t)$ is used to denote the external boundaries of the material control volume $\Omega(t)$. The normal vector $\mathbf{n}$ points outward from $\Omega(t)$.

As often for the case of turbulent flows, the numerical treatment of the convective transport term underlying the material derivative operator in Eqs. (7)–(9) is crucial, and the choices made are known to affect the accuracy, conservation, and stability properties of the resulting algorithm. This is especially true for multiphase flow and is at the core of the current method. The related challenges are discussed in the following section, along with an introduction to the proposed strategy.

2.3. Towards a Lagrange-Remap discretization

In this section, a model equation for purely convective transport in a solenoidal velocity field is considered and used to illustrate the challenges faced when discretizing the convective term in the marker, density, and momentum transport equations. The variable φ is used interchangeably to denote the volume fraction (f), the density (ρ), the α -component of the momentum (ρu_α) or velocity (u_α) vectors.

Discretizing the convective transport term in the conservation equations gives rise to two fundamental issues. On the one hand, considering the advection equation

$$\frac{\partial\varphi}{\partial t} + \mathbf{u} \cdot \nabla\varphi = 0, \quad (10)$$

it is straightforward to see that, provided the fields $\mathbf{u}$ and φ are smooth, φ is constant along the characteristics and should therefore maintain compact support at all time. Discretely, such a property is also desirable, for the creation of non-physical extrema in the updated field is detrimental to both accuracy and stability of the method. A monotonicity preserving discretization of the convective transport term is therefore highly desirable for stable computations.

On the other hand, if the velocity field $\mathbf{u}$ is solenoidal, which is the case in the current problem, Eq. (10) may also be recast in a conservative form:

$$\frac{\partial\varphi}{\partial t} + \nabla \cdot (\varphi\mathbf{u}) = 0. \quad (11)$$

For finite difference and finite volume formulations, discrete primary and secondary conservation properties have been known to be suitable for accurate and stable computations [21,22]. Therefore, conserving momentum and kinetic energy in the inviscid limit of the Navier–Stokes equations is also suitable for accurate two-phase flow computations.

In general, guaranteeing either conservation or monotonicity preservation alone is easy, but deriving a scheme without excessive numerical dissipation that satisfies both properties simultaneously is very challenging. Therefore, the focus of the proposed work is the discretization of the conservative form of the Navier–Stokes equations which maintains compact support for the density, momentum, and velocity. While sharp interface formulations guaranteeing such properties for the volume fraction (or equivalently for incompressible fluids, the density) have previously been developed, this has to the best of our knowledge never been done for either the momentum prediction or the momentum rescaling leading to the updated velocity field.

Here, before considering the momentum equation, we will first discuss the volume fraction equation. For readability, we also introduce for any control volume $\Omega(t)$ the following notation

$$\varphi_\Omega(t) = \frac{\int_{\Omega(t)} \varphi(\mathbf{x}, t) dV}{|\Omega(t)|}, \quad (12)$$

which makes use of the $|\Omega(t)|$ notation, previously defined as the volume of $\Omega(t)$. φ_Ω therefore denotes the volume averaged value of φ . In the Lagrange-Remap context underlying the material derivatives used in Eqs. (7)–(9), conservation and monotonicity preservation become apparent when decomposing the transport equations as follows. The integral form of the volume fraction transport equation (Eq. (7)) may be rewritten for an incompressible flow with no mass transfer across the interface

$$\frac{d}{dt}(f_\Omega(t)|\Omega(t)|) = f_\Omega(t) \frac{d}{dt}|\Omega(t)| + |\Omega(t)| \frac{d}{dt}f_\Omega(t) = 0. \quad (13)$$

From Eq. (8), the first term in the right-hand side of Eq. (13) is null, leading to

$$\frac{d}{dt}f_\Omega(t) = 0, \quad (14)$$

which suggests that for any material control volume, the averaged indicator function value remains constant. In the limit $|\Omega(t)| \rightarrow 0$, this of course leads back to the indicator function being constant along the characteristics. From a finite volume standpoint, this is a good starting point to discretize Eq. (7) without creating new extrema. However, the main challenge is to simultaneously and discretely satisfy Eq. (8), which leads to local volume conservation. If the discretization solves Eq. (14), it is by construction monotonicity preserving. To satisfy the conservation property, the problem faced at this point can be stated in the following task: deform a given mesh with the fluid velocity while maintaining a constant volume for each element.

As pointed out above, this task is rarely accomplished, even in methods using dimensional splitting. Notable exceptions are the dimensional splitting algorithms presented in [23,24] in two and three dimensions, respectively, and the two-dimensional unsplit method introduced in [25]. In the unsplit method presented in [26], Eq. (14) is solved. It is therefore monotonicity preserving, but not conservative. The conservation error, however, was found to be of small amplitude and therefore not to affect the overall accuracy of the method. The foremost important step here is the formulation of a monotonicity preserving scheme for the momentum transport. In the proposed method, the key idea for this is to apply a similar methodology to each component α of momentum defined over a control volume $\Omega_\alpha(t)$. The discretization of the convective transport term then follows directly from a consistent discretization of

$$\begin{cases} \frac{d}{dt}(\rho u_\alpha)_{\Omega_\alpha}(t) = 0, \\ \frac{d}{dt}(\rho)_{\Omega_\alpha}(t) = 0, \end{cases} \quad (15)$$

which will be shown to guarantee the maximum principle for both momentum and velocity while retaining a second order accurate interface representation. Specific treatments for the viscous and capillary effects follow those from existing work [14,16], but the correction step still needs to be modified consistently with the discretization of Eq. (15) in order to guarantee momentum conservation.

3. A monotonicity preserving sharp interface discretization for convective transport

A fundamental improvement of the proposed method lies in the geometric treatment of the convective transport term in the governing equations. This is the focus of this section, which presents two geometric Lagrange-Remap advection schemes that are later applied to the continuity and momentum equations. A high-level description is adopted here. Implementation details easily follow from the work presented in [26] where a similar discretization is used to solve the volume fraction transport equation. Particular attention is put on the formalism in order to bridge the gap with the analytical equations presented in Section 2 and to facilitate the stability and conservation analysis that will be presented in Sections 3.3 and 3.4.

In order to define a proper discretization of the convective transport term in the momentum equation along with an appropriate rescaling to update the velocity field, the following model transport equation is considered

$$\frac{d}{dt}\varphi_\Omega(t) = 0, \quad (16)$$

which is to be solved for the marker, density, and momentum variables using a Lagrange-Remap method for every mesh cell Ω over which they are defined. To do so, Lagrange-Remap methods proceed as follows. First, the equations are advanced in a Lagrangian coordinate system. This is trivial in the integral form considered, which neglects all but convective transport, since the right-hand side in Eq. (16) is null. Second, a change of coordinate system (from Lagrangian to Eulerian), or remapping, is carried out to transfer the information onto the Eulerian computational mesh. This can be done by following the material control volume $\Omega(t)$ backward or forward in time along the characteristics. Every point $\mathbf{x}(t)$ in $\Omega(t)$ is therefore transported according to

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{u}(\mathbf{x}, t). \quad (17)$$

This leads to two different temporal discretizations that are presented below, which will be called backward and forward time advancement schemes. In the proposed framework, due to the sharp representation inherent to the problem at hand, the remapping consists of a series of polyhedral operations, described in detail in [26]. Since all faces of the control volumes are assumed to be piecewise planar, this limits the accuracy to second order, which is sufficient for a large range of problems. A description of this procedure will be provided in Section 3.1.

In addition to the remapping step, in order to retain the sharp representation adopted in this work, an estimate of the interface front has to be determined within each control volume prior to the geometric interpolation. For the sake of completeness, we also mention that, even though the interface reconstruction is sufficient to obtain formal second order accuracy for piecewise constant functions (such as the marker function f or the density ρ), a linear reconstruction is required for the momentum/velocity transport to be formally second order accurate. While easy to include in the geometric transport, it will be shown in Section 3.2) that simultaneously maintaining second order accuracy for the viscous term and the pressure projection equation while guaranteeing conservation raises a fundamental challenge in the scope of the one-fluid approach adopted in this work. Since the geometric transport will ultimately be used only over a very narrow band centered around the interface, this is not expected to affect the results. Further details are included in Section 3.2.

Interestingly, the Lagrange-Remap discretizations proposed here may be expressed in the form of a geometric upwind finite volume scheme. The choice is made to focus first on the Lagrange-Remap perspective for clarity, but details on this equivalence are provided in Section 3.3, in which the conservation error is also identified. In addition, the particular form of Eq. (16), which is used to achieve a non-oscillatory scheme as motivated in Section 2), is formally proven in Section 3.4) to be non-oscillatory for density and momentum as well as for the velocity resulting from the momentum rescaling.

3.1. Geometric transport

Practically, the most challenging task in the proposed algorithm is the remapping step of the Lagrange-Remap procedure. It consists of transferring the information from a computational mesh (referred to as Eulerian mesh) to a Lagrangian element or from a set of Lagrangian elements to a single cell of the Eulerian mesh. These two operations are involved in the backward and forward Euler time advancements, respectively.

Although not presented in depth here, the deformation of the mesh cells in time to solve Eq. (16) is critical to the stability and conservation properties of the transport. This part of the algorithm is described in detail in [26], in which, by advecting the mesh vertices and using a properly defined triangulation of each element, the Lagrangian mesh is guaranteed to define a non-overlapping partition of the computational domain. Given such a decomposition, the updated values result from a volume averaged sum of elementary integrals computed over the intersections of Eulerian and Lagrangian elements. Since all elements are linear, the domains over which the integrals are computed are polyhedral intersections, which are computed using an in-house polyhedral library described in detail in [26].

3.1.1. Backward Euler

Given a computational mesh at time $n + 1$, one possible discretization of Eq. (16) follows from tracking each element $\Omega(t^{n+1}) = \mathcal{E}_i^{n+1}$ of the mesh (i is the element index) backward in time with the fluid velocity. This leads to the same material control volume, but now deformed at time n , and denoted $\Omega(t^n) = \mathcal{L}_i^n$. Since the field φ is known at time n in each element $\mathcal{E}_j^n$, the mean value φ_i^{n+1} over $\mathcal{L}_i^n$ is simply given by the formula

$$\varphi_{\mathcal{L}_i^n} = |\mathcal{L}_i^n|^{-1} \sum_j \int_{\mathcal{L}_i^n \cap \mathcal{E}_j^n} \varphi(\mathbf{x}, t^n) dV, \quad (18)$$

where the elementary integrals require the knowledge of the estimated distribution of φ at time n within each Eulerian cell, as shown in Fig. 1(a) for the interface reconstruction. How this reconstruction is computed will be presented in the following section.

According to Eq. (16), this value is exactly the mean value of φ over $\mathcal{E}_i^{n+1}$, which formally reads

$$\varphi_{\mathcal{E}_i^{n+1}} = \varphi_{\mathcal{L}_i^n}. \quad (19)$$

The field φ is therefore updated using Eq. (19).

3.1.2. Forward Euler

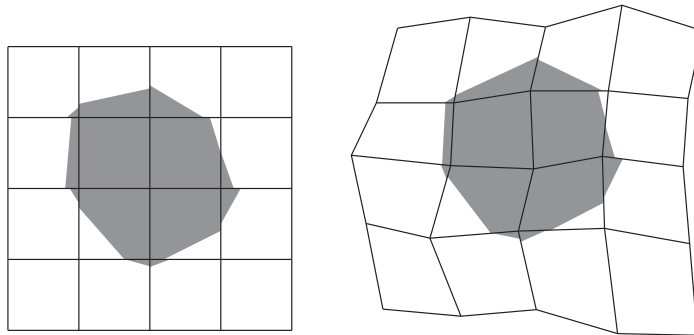
Alternatively, one could track each element $\Omega(t^n) = \mathcal{E}_j^n$ of the mesh at time n forward in time, resulting in a set of Lagrangian elements $\Omega(t^{n+1}) = \mathcal{L}_j^{n+1}$ covering the computational domain at time $n + 1$. The averaged φ -value over each Lagrangian element is indeed known from the evolution Eq. (16), approximated here with

$$\varphi_{\mathcal{L}_j^{n+1}} = \varphi_{\mathcal{E}_j^n}, \quad (20)$$

and which simply states that the volume averaged φ over $\mathcal{L}_j^{n+1}$ is the same as for $\mathcal{E}_j^n$. Therefore, this can be used to update the volume averaged value of φ for each element $\mathcal{E}_i^{n+1}$

$$\varphi_{\mathcal{E}_i^{n+1}} = |\mathcal{E}_i^{n+1}|^{-1} \sum_j \int_{\mathcal{E}_i^{n+1} \cap \mathcal{L}_j^{n+1}} \varphi(\mathbf{x}, t^{n+1}) dV, \quad (21)$$

where the φ -distribution is now required at time $n + 1$ over each element $\mathcal{L}_j^{n+1}$ as shown in Fig. 1(b) for the interface reconstruction. How this reconstruction is built is described next.



(a) Backward Euler, time n . (b) Forward Euler, time $n + 1$.

Fig. 1. Required interface representation before remapping.

3.2. Reconstructions

The integrands in Eqs. (18) and (21) have to be treated with care. A simple approach would, for example, consist of approximating φ by a constant over each element $\mathcal{E}_j^n$ or $\mathcal{L}_j^{n+1}$ for the backward or forward transport, respectively. However, for $\varphi = f$ for example, such an approach would rapidly spread the interface over a few mesh cells. To address this, Volume-of-Fluid methods limit the numerical diffusion by locally reconstructing the interface. The particular method used for this reconstruction in the present study will be briefly described in the first part of this section.

In addition, while a piecewise linear interface reconstruction is sufficient to achieve second order accuracy when transporting a piecewise constant function (such as the indicator function or the density field), this is not sufficient for the momentum equation. This can easily be included in the convective transport in a stable and conservative way, but for non-unity density ratio, maintaining second order accuracy throughout the flow solver becomes a challenge. This point is discussed in the second part of this section.

3.2.1. Interface

A Piecewise Linear Interface Calculation (PLIC, [27]) method is used to locally reconstruct the interface. The procedure follows the algorithm proposed in the original Coupled Level Set and Volume-of-Fluid method [28], briefly described here. Given an Eulerian or Lagrangian cell ($\mathcal{E}_i^n$ or $\mathcal{L}_i^{n+1}$, respectively), an initial guess of the interface plane is first obtained by fitting (in a least square sense) a linear function to the Level Set field neighboring the cell at the same time level (n or $n + 1$, respectively). In order for the reconstruction to be conservative, this plane is then translated until the local volume fraction value is matched. Details can be found in [26].

3.2.2. Velocity

Formally achieving second order accuracy in solving Eq. (16) for the momentum requires a piecewise linear reconstruction of the velocity field before evaluating each elementary integral in Eqs. (18) and (21). This reconstruction should remain conservative without introducing any new extrema for the velocity.

Assuming the velocity gradient is continuous across the interface (this is typically assumed in one-fluid formulations), a linear reconstruction can be achieved for every α -momentum control volume $\Omega_{\alpha,i}$ (i being the cell index) by fitting the following function

$$u_\alpha(\mathbf{x}, t) = \hat{U}_{\alpha,i} + \mathbf{G}_\alpha \cdot (\mathbf{x} - \hat{\mathbf{x}}_{\alpha,i}), \quad \forall \mathbf{x} \in \Omega_{\alpha,i}, \quad (22)$$

where $\mathbf{G}_\alpha$ denotes the local estimate of the spatial velocity gradient, and the following notations were used

$$\hat{U}_{\alpha,i} = \frac{\int_{\Omega_{\alpha,i}} \rho(\mathbf{x}, t) u_\alpha(\mathbf{x}, t) dV}{\int_{\Omega_{\alpha,i}} \rho(\mathbf{x}, t) dV} \quad \text{and} \quad \hat{\mathbf{x}}_{\alpha,i} = \frac{\int_{\Omega_{\alpha,i}} \mathbf{x} \rho(\mathbf{x}, t) dV}{\int_{\Omega_{\alpha,i}} \rho(\mathbf{x}, t) dV}, \quad (23)$$

which denote the rescaled momentum (or mass averaged velocity) and the center of mass of $\Omega_{\alpha,i}$, respectively. An appropriate slope limiter should also be used in order to prevent new extrema. Note also that the only unknown in Eq. (22) is the velocity gradient $\mathbf{G}_\alpha$, since for the reconstruction to conserve momentum, $\hat{U}_{\alpha,i}$ needs to be set to the velocity value in cell $\Omega_{\alpha,i}$.

This remark raises the following point: the velocity value $\hat{U}_\alpha$ stored for every cell is mass centered. Indeed, if the velocity field were linear, $\hat{U}_{\alpha,i}$ would be equal to the value of the reconstructed velocity u_α evaluated at the center of mass $\hat{\mathbf{x}}_\alpha$, not at the geometric cell center. When discretely solving for the characteristics Eq. (17), on the other hand, the velocity interpolation (trilinear in the current implementation) typically uses the cell centered values

$$\bar{U}_{\alpha,i} = \frac{\int_{\Omega_{\alpha,i}} u_\alpha(\mathbf{x}, t) dV}{\int_{\Omega_{\alpha,i}} dV} \quad (24)$$

as do the discretizations of the viscous term and the pressure projection equation in order to achieve second order accuracy. Of course, mass and volume centered velocities are not expected to match, and accounting for this difference without affecting the conservation properties of the scheme while maintaining second order accuracy turns out to be a challenging task.

Instead, the choice is made to limit the formal accuracy of the geometric discretization of the convective transport to first order by setting $\mathbf{G}_\alpha = \mathbf{0}$, but to maintain the interface treatment second order (PLIC). To reduce the numerical dissipation inherent in this first order approximation, the use of the geometric transport is limited to a narrow band of cells located in the interface vicinity, and a second order centered discretization [29] is used in other regions of the flow.

3.3. Finite volume equivalents and conservation error

The geometric update formulas Eqs. (18) and (21) may both be rewritten in a finite volume sense. Various definitions are first introduced in this section, which are then used along with the notations defined in [21] to formulate discrete finite volume equivalents of the backward and forward Lagrange-Remap transport. These equations will then be used to interpret the origin of the conservation error and its link with the governing equations presented in Section 2.

These notations will ultimately be used in Section 4 to derive the full discretization of the Navier–Stokes equations.

3.3.1. Flux definition

In order to prove the proposed Lagrange-Remap transport to be equivalent to a finite volume scheme, it is intuitive to make use of the volumes fluxed by the forward or backward transport of the faces adjacent to each control volume. These fluxes have to be constructed consistently with how the mesh elements are deformed in the backward and forward Lagrange-Remap integrations.

Backward Euler. In order to define the fluxes resulting from the backward time integration, the transformation of a face $\mathcal{A}_{\mathcal{E}}^{n+1}$ of the Eulerian mesh at time $n + 1$ into a face $\mathcal{A}_{\mathcal{L}}^n$ of the Lagrangian mesh at time n is represented in Fig. 2. Because the flux evaluation will rely on the interface reconstruction available at time n on the Eulerian mesh (also represented in Fig. 2), the volume bounded by the planar faces $\mathcal{A}_{\mathcal{L}}^n$ and $\mathcal{A}_{\mathcal{E}}^{n+1}$ is denoted $\mathcal{V}^n$ and represented by the red area in Fig. 2.

The definition of $\mathcal{V}^n$, however, is not yet complete. As seen in 2D in Fig. 2, $\mathcal{V}^n$ is bounded by two planar faces $\mathcal{A}_{\mathcal{L}}^n$ and $\mathcal{A}_{\mathcal{E}}^{n+1}$ and two streamlines. These streamlines, purposely represented with curved lines in Fig. 2, do not actually need to be defined as long as they also bound the neighboring fluxed volumes. This is illustrated in Fig. 2, where a second fluxed volume depicted in green and neighboring the red one is represented. As long as the approximation of the streamline is shared by both volumes (no gaps and no overlaps), the effect on the updated values is unchanged.

Forward Euler. A similar flux definition follows for the forward time integration. The fluxed volume, now delimited by two faces $\mathcal{A}_{\mathcal{E}}^n$ and $\mathcal{A}_{\mathcal{L}}^{n+1}$, is denoted $\mathcal{V}^{n+1}$. The $n + 1$ superscript is used because the fluxes will be evaluated based on the interface reconstruction at time $n + 1$ on the Lagrangian mesh, also represented on Fig. 3. As pointed out earlier for the backward fluxed volumes, the updated values do not depend on the approximation of the streamlines as long as the neighboring fluxed volumes are contiguous.

One additional remark related to the sign of the fluxed volume needs to be made. In the backward integration configuration pictured in Fig. 2, the red area corresponds to a volume flowing into the grayed cell (to the right of face $\mathcal{A}_{\mathcal{E}}^{n+1}$). If face $\mathcal{A}_{\mathcal{L}}^n$ were to the right of this face, on the other hand, the volume $\mathcal{V}^n$ should be accounted as negative (flowing out of the grayed cell). Similarly, in the forward integration configuration depicted in Fig. 3, the red area also corresponds to a flux flowing into the grayed cell (to the right of face $\mathcal{A}_{\mathcal{E}}^n$). This time, however, $\mathcal{V}^{n+1}$ should be accounted as flowing out of the grayed cell if face $\mathcal{A}_{\mathcal{L}}^{n+1}$ were located to the left of face $\mathcal{A}_{\mathcal{E}}^n$.

In both schemes, a mixture of inflowing and outflowing fluxes might also occur simultaneously. This subtlety therefore needs to be accounted for in the notation. This is done by supplementing the superscripts on the $\mathcal{V}$ notations with an additional index ϵ . ϵ denotes the fluxes inflowing ($\epsilon = +$) and outflowing ($\epsilon = -$) the cell located on the positive side of the Eulerian face. Since the analysis performed here is for structured Cartesian meshes (similar results are valid for unstructured meshes), each face $\mathcal{A}_{\mathcal{E}}$ is perpendicular to one axis β of the coordinate system: the positive side of a face $\mathcal{A}_{\mathcal{E}}$ is therefore defined as the side pointing in the direction of positive abscissas. An additional superscript β is included to differentiate the faces.

For the backward integration, using these definitions, the flux decomposition at a face $\mathcal{A}_{\mathcal{E}}^{\beta, n+1}$ perpendicular to the β -axis reads

$$\mathcal{V}^{\beta, n} = \cup_{\epsilon \in \{+, -\}} \mathcal{V}^{\epsilon \beta, n}. \quad (25)$$

One can therefore define the following effective velocities

$$\mathbf{v}_{\epsilon \beta}^n = \epsilon \frac{|\mathcal{V}^{\epsilon \beta, n}|}{\Delta t |\mathcal{A}_{\mathcal{E}}^{n+1}|}, \quad (26)$$

where,

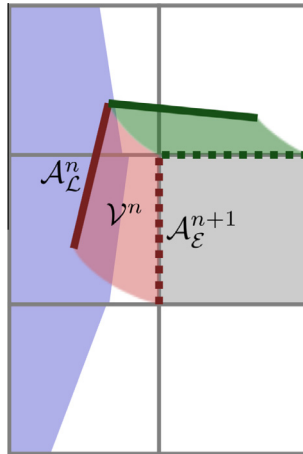


Fig. 2. Definition of the volume swept during the backward transformation of an Eulerian face $\mathcal{A}_{\mathcal{E}}^{n+1}$ (dashed red line) into a Lagrangian face $\mathcal{A}_{\mathcal{L}}^n$ (solid red line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

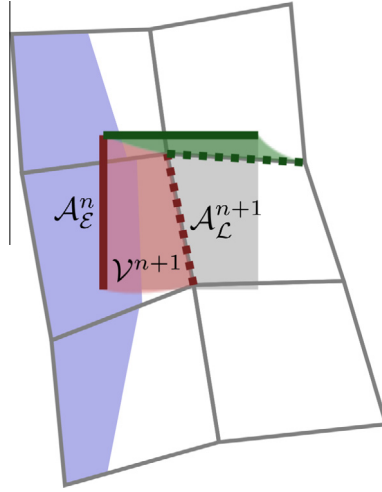


Fig. 3. Definition of the volume swept during the forward transformation of an Eulerian face $\mathcal{A}_E^n$ (solid red line) into a Lagrangian face $\mathcal{A}_L^{n+1}$ (dashed red line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$\begin{cases} |\mathcal{V}^{\epsilon\beta,n}| = \int_{\mathcal{V}^{\epsilon\beta,n}} d\mathcal{V}, \\ |\mathcal{A}_E^{n+1}| = \int_{\mathcal{A}_E^{n+1}} d\mathcal{A}. \end{cases} \quad (27)$$

Similar definitions follow for the forward integration, one simply needs to interchange the n and $n+1$ indices.

Using the effective velocities, the φ -fluxes at each face may finally be expressed for the backward transport

$$\tilde{\varphi}^{\epsilon\beta,n} v_{\epsilon\beta}^n \mathcal{A}^{\beta,n+1} \Delta t, \quad (28)$$

and the forward transport

$$\tilde{\varphi}^{\epsilon\beta,n+1} v_{\epsilon\beta}^{n+1} \mathcal{A}^{\beta,n}. \quad (29)$$

The following notation was used,

$$\tilde{\varphi}^{\epsilon\beta,m} = |\mathcal{V}^{\epsilon\beta,m}|^{-1} \int_{\mathcal{V}^{\epsilon\beta,m}} \varphi(\mathbf{x}, t^m) d\mathcal{V}, \quad (30)$$

$m \in \llbracket n, n+1 \rrbracket$. This definition can also be used to write the density averaged flux value

$$\hat{\varphi}^{\epsilon\beta,m} = \frac{\tilde{\varphi}^{\epsilon\beta,m}}{\tilde{\rho}^{\epsilon\beta,m}}, \quad (31)$$

which will later be used in the momentum advection equation.

3.3.2. Transport equations

The following discrete geometric conservation law [30] is central to the equivalence between the Lagrange-Remap and Eulerian versions of the proposed method

$$\begin{cases} \frac{|\mathcal{E}^{n+1}| - |\mathcal{E}^n|}{\Delta t} = |\mathcal{E}^{n+1}| \frac{\delta_1 v_\beta^n}{\delta_1 x_\beta}, \\ \frac{|\mathcal{L}^{n+1}| - |\mathcal{L}^n|}{\Delta t} = |\mathcal{E}^n| \frac{\delta_1 v_\beta^{n+1}}{\delta_1 x_\beta}, \end{cases} \quad (32)$$

where $v_\beta^m = \sum_{\epsilon} v_{\epsilon\beta}^m$ ($m \in \llbracket n, n+1 \rrbracket$). $\delta_1 \cdot / \delta_1 x_\beta$ denotes the second order divergence operator defined in [21]. At this point, it is worth noting that the upper indices n and $n+1$ are merely an abusive notation, and are not intended to denote an estimate of the velocity field at time n and $n+1$, respectively. Eq. (32) provided indeed two admissible second order discretizations of Eq. (8) since it can be proven that the proposed geometric advection scheme deforms each control volume $\mathcal{E}^n$ and $\mathcal{E}^{n+1}$ exactly in a linear velocity field [26].

The connection between the effective velocities v_β defined at each face of a mass control volume and the flow velocity values u_β given on a staggered mesh [21] is obvious. A fundamental difference, however, is that, while the discretization of the continuity equation (6) results in the updated velocity field satisfying

$$\frac{\delta_1 u_\beta}{\delta_1 x_\beta} = 0, \quad (33)$$

the same is true for v_β^m ($m \in \llbracket n, n+1 \rrbracket$) only for linear and solenoidal velocity fields [26]. Since for complex turbulent flows the velocity is all but linear, this is expected to affect the conservation properties of the proposed scheme; the resulting error, however, remains very small.

The updated volume averaged values φ^B and φ^F for the backward and forward schemes are given by

$$\begin{cases} \frac{\varphi^B - \varphi^n}{\Delta t} + \frac{\delta_1 \widetilde{\varphi}^{\epsilon\beta,n} v_{\epsilon\beta}^n}{\delta_1 x_\beta} = \varphi^B \frac{\delta_1 v_\beta^n}{\delta_1 x_\beta}, \\ \frac{\varphi^F - \varphi^n}{\Delta t} + \frac{\delta_1 \widetilde{\varphi}^{\epsilon\beta,n+1} v_{\epsilon\beta}^{n+1}}{\delta_1 x_\beta} = \varphi^n \frac{\delta_1 v_\beta^{n+1}}{\delta_1 x_\beta}, \end{cases} \quad (34)$$

where Einstein notation is used on $\epsilon \in \{-, +\}$ and $\beta \in \llbracket 1, d \rrbracket$, d denotes the dimension of the problem, and B and F stand for backward and forward, respectively. An immediate conclusion stemming from Eq. (34) is that both backward and forward transport are constant preserving. This is reassuring since they are intended to transport an indicator function. Also immediate is the implicit nature of the backward step (the right-hand side involves the updated value) and the explicit nature of the forward step. It will be shown that this discretization is not only constant preserving but also monotonicity preserving.

In spite of the previous Eq. (34) being sufficient to fully characterize the volume fraction, density, and momentum convective transport, they are decomposed further (here only for the forward scheme) using the density weighted averaging operator Eq. (31)

$$\begin{cases} \frac{\rho_\alpha^F - \rho_\alpha^n}{\Delta t} + \frac{\delta_1 \widetilde{\rho}_\alpha^{\epsilon\beta} v_{\epsilon\beta}^n}{\delta_1 x_\beta} = \rho_\alpha^n \frac{\delta_1 v_\beta^n}{\delta_1 x_\beta}, \\ \frac{g_\alpha^F - g_\alpha^n}{\Delta t} + \frac{\delta_1 \widetilde{\rho}_\alpha^{\epsilon\beta} u_\alpha^{\epsilon\beta} v_{\epsilon\beta}^n}{\delta_1 x_\beta} = g_\alpha^n \frac{\delta_1 v_\beta^n}{\delta_1 x_\beta}, \end{cases} \quad (35)$$

where g denotes the momentum variable and the $n+1$ indices on the flux related values are dropped for clarity. Note also that α simply denotes with respect to which control volumes the fluxes are defined. This is required since the variables are staggered in space.

Eq. (35) is the forward euler discretization of Eq. (15). The consequences of solving an equation of the form of Eq. (15) in place of Eq. (7) is the appearance of a source term in the right-hand side of Eq. (35), which introduces a conservation error. This term, however, is required for the scheme to be non-oscillatory as will be shown in the next section. In addition, the divergence of the effective velocity is discretely zero for solenoidal and linear velocity fields [26]. Therefore, this term is to be of order higher than second (though this is challenging to prove analytically); its amplitude, as shown in the numerical experiments [26] remains small.

3.4. Stability analysis

While the Eulerian formulation described above is useful for understanding the origin of the conservation error, the Lagrange-Remap description underlying Eqs. (18) and (21) is essential to show that the updated values satisfy the maximum principle inherent in the advection equation. The analysis carried here focuses on the backward transport but can be similarly performed for the forward scheme.

In order to prove that the maximum principle discretely holds, Eq. (18) for a given φ -control volume i is now expressed as follows

$$\varphi_i^{n+1} = \sum_j \omega_{ij}^L \langle \varphi \rangle_{ij}^L + \omega_{ij}^G \langle \varphi \rangle_{ij}^G, \quad (36)$$

where j indicates the cells neighboring i . The weights are defined as

$$\begin{cases} \omega_{ij}^L = |\mathcal{L}_i^n|^{-1} \int_{\mathcal{L}_i^n \cap \mathcal{E}_j^n} f(\mathbf{x}, t^n) dV, \\ \omega_{ij}^G = |\mathcal{L}_i^n|^{-1} \int_{\mathcal{L}_i^n \cap \mathcal{E}_j^n} [1 - f(\mathbf{x}, t^n)] dV, \end{cases} \quad (37)$$

and the abscissas as

$$\begin{cases} \langle \varphi \rangle_{ij}^L = \frac{\int_{\mathcal{L}_i^n \cap \mathcal{E}_j^n} f(\mathbf{x}, t^n) \varphi(\mathbf{x}, t^n) dV}{\int_{\mathcal{L}_i^n \cap \mathcal{E}_j^n} f(\mathbf{x}, t^n) dV}, \\ \langle \varphi \rangle_{ij}^G = \frac{\int_{\mathcal{L}_i^n \cap \mathcal{E}_j^n} [1 - f(\mathbf{x}, t^n)] \varphi(\mathbf{x}, t^n) dV}{\int_{\mathcal{L}_i^n \cap \mathcal{E}_j^n} [1 - f(\mathbf{x}, t^n)] dV}. \end{cases} \quad (38)$$

The next step consists of showing that the weights define a convex sum, while the abscissas are bounded by the extremas of the reconstructed φ -field.

From Eq. (37), it is trivial to check that provided the sets $\{\mathcal{L}_i^n\}$ and $\{\mathcal{E}_i^n\}$ both define a non-overlapping partition of the computational domain, the following properties hold

$$\forall (i, j), \quad 0 \leq \omega_{ij}^{L,n} \leq 1 \quad \text{and} \quad 0 \leq \omega_{ij}^{G,n} \leq 1, \quad (39)$$

and

$$\forall i, \sum_j (\omega_{ij}^{L,n} + \omega_{ij}^{G,n}) = 1. \quad (40)$$

As pointed out, this property obviously holds only for properly defined meshes, as emphasized in [26].

Since the abscissas are by definition bounded by the extremas of the reconstructed field $\varphi(\mathbf{x}, t^n)$, guaranteed to be bounded by the neighboring cell-averaged values φ_j^n , one can write

$$\min_{j|\omega_{ij} \neq 0} \varphi_j^n \leq \varphi_i^{n+1} \leq \max_{j|\omega_{ij} \neq 0} \varphi_j^n. \quad (41)$$

Since the Lagrange-Remap transport is used for the volume fraction, density, and momentum variables, all of them are guaranteed to remain bounded by the local extrema. For the momentum as mentioned above, this is conditioned on not violating the maximum principle in the reconstruction step, which is trivial if a piecewise constant velocity is used.

Satisfaction of the maximum principle for the velocity remains to be proved. The velocity is updated by rescaling the momentum

$$(u_x)_i^* = \frac{(\mathbf{g}_x)_i^*}{(\rho_x)_{i+1}^{n+1}}, \quad (42)$$

where as previously, ρ_x is defined with respect to the x -momentum control volume.

The proof follows easily. Defining density averaged weights

$$\hat{\omega}_{ij}^L = \frac{\rho^L \omega_{ij}^L}{\rho_{ij}^{n+1}} \quad \text{and} \quad \hat{\omega}_{ij}^G = \frac{\rho^G \omega_{ij}^G}{\rho_{ij}^{n+1}}, \quad (43)$$

which also define a convex sum, and using the fact that the density is constant in each phase, one can rewrite the rescaling equation (42) as

$$(u_x)_i^* = \sum_j \hat{\omega}_{ij}^L \langle u_x \rangle_{ij}^L + \hat{\omega}_{ij}^G \langle u_x \rangle_{ij}^G, \quad (44)$$

where the updated velocity can also be seen to satisfy the maximum principle.

3.5. Equivalent velocity transport equation

At this point, it is worth remembering that the velocity field is not directly transported, but instead computed from the primitive variables. From Eq. (35) however, for the sake of completeness, we report the discrete transport equation for the velocity which may be written following the notations in [21]

$$\bar{\rho}_x^{1t} \left(\frac{u_x^* - u_x^n}{\Delta t} + \frac{\bar{\rho}_x^{\epsilon\beta} v_{\epsilon\beta}^{1\beta}}{\bar{\rho}_x^{1t}} \frac{\delta_1 \widehat{u}_x^{\epsilon\beta}}{\delta_1 x_\beta} \right) = \sum_\epsilon \left[\frac{u_x^* + u_x^n}{2} - \widehat{u}_x^{\epsilon\beta 1\beta} \right] \frac{\delta_1 \tilde{\rho}_x^{\epsilon\beta} v_{\epsilon\beta}^\alpha}{\delta_1 x_\beta} + \sum_\epsilon \rho_x^n \frac{u_x^n - u_x^*}{2} \frac{\delta_1 v_{\epsilon\beta}^\alpha}{\delta_1 x_\beta}. \quad (45)$$

4. A stable and close-to-conservative pressure-correction method

Section 3 motivated and presented in details the discretization of the convective transport term in the conservative form of the two-phase Navier–Stokes equations. Using the notations defined in Section 3.3, the focus is now on the full prediction-correction algorithm. The classical treatments of the viscous and surface tension terms used in this work are first presented in the scope of the Ghost Fluid method [15]. Second, the time integration scheme is presented in detail, with particular emphasis on the proposed pressure correction step.

4.1. Ghost fluid method

Given a one-dimensional mesh crossed by an interface at x_i between two vertices x_i and x_{i+1} , a technique often encountered in two-phase flow computations is the GFM. It is assumed here that the liquid phase lies on the right of the interface ($x \geq x_i$). Introduced for second order partial differential equations to improve interface boundary condition enforcement, it consists in making use of the two jump conditions

$$[\xi]_i, \quad [\chi \frac{\partial \xi}{\partial x}]_i \quad (46)$$

to increase the formal accuracy of the discretizations of

$$\chi \frac{\partial \xi}{\partial x} \Big|_{i+1/2}^G, \quad \chi \frac{\partial \xi}{\partial x} \Big|_{i+1/2}^L. \quad (47)$$

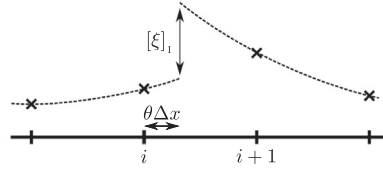


Fig. 4. Definition of the Ghost Fluid parameters.

The subscript on the jump operator $[\![\cdot]\!]$ simply denote where the jumps are evaluated: at the interface (I), or at a mesh nodes (i or $i + 1$). χ may refer to the specific volume if the GFM is applied to the pressure term or to the dynamic viscosity if applied to the viscous term (see Fig. 4).

The following identities may be shown to hold to second order

$$\begin{cases} \chi^L [\![\xi]\!]_i = \chi^\star [\![\xi]\!]_I - \frac{\chi^\star}{\chi^G} \theta \Delta x [\![\chi \frac{\partial \xi}{\partial x}]\!]_I + \frac{\chi^\star [\![\chi]\!]_I}{\chi^G} \theta (\xi_{i+1} - \xi_i), \\ \chi^G [\![\xi]\!]_{i+1} = \chi^\star [\![\xi]\!]_I + \frac{\chi^\star}{\chi^G} (1 - \theta) \Delta x [\![\chi \frac{\partial \xi}{\partial x}]\!]_I - \frac{\chi^\star [\![\chi]\!]_I}{\chi^G} (1 - \theta) (\xi_{i+1} - \xi_i), \end{cases} \quad (48)$$

where ξ_i is defined in the left (gaseous, G) phase and ξ_{i+1} in the right (liquid, L) phase, and the following harmonic averaging is used

$$\frac{1}{\chi^\star} = \frac{\theta}{\chi^G} + \frac{1 - \theta}{\chi^L}, \quad (49)$$

and

$$\theta = \frac{x_I - x_i}{x_{i+1} - x_i}. \quad (50)$$

This may be shown to ultimately lead to the following first order discretizations

$$\begin{cases} \chi \frac{\partial \xi}{\partial x} \Big|_{i+1/2}^G = \chi^\star \frac{\xi_{i+1} - \xi_i}{\Delta x} - \frac{\chi^\star [\![\xi]\!]_I}{\Delta x} + \frac{\chi^\star}{\chi^G} \theta [\![\chi \frac{\partial \xi}{\partial x}]\!]_I, \\ \chi \frac{\partial \xi}{\partial x} \Big|_{i+1/2}^L = \chi^\star \frac{\xi_{i+1} - \xi_i}{\Delta x} - \frac{\chi^\star [\![\xi]\!]_I}{\Delta x} - \frac{\chi^\star}{\chi^L} (1 - \theta) [\![\chi \frac{\partial \xi}{\partial x}]\!]_I. \end{cases} \quad (51)$$

4.1.1. Viscous stress

In the presence of an interface, the discretization of the viscous term needs specific treatment. In the Continuum Surface Force method [12], the dynamic viscosity jump is spread over a finite number of cells. Alternatively, volumetric averaging of the viscosity is widely used.

Another approach [14] consists in using a harmonic averaging of the dynamic viscosity in the vicinity of the interface. In the context of the GFM, this may be justified in a one-dimensional case as follows. Setting $\xi = u$ and $\chi = \mu$, for a continuous velocity field and assuming

$$\left[\mu \frac{\partial u}{\partial x} \right]_I = 0, \quad (52)$$

the set of equations previously derived Eq. (51) naturally leads to the harmonic averaging introduced in [14]. This type of averaging is however valid only when the interface is perpendicular to the direction of differentiation, but its extension to two- and three-dimensional flows was still found to be more accurate than arithmetic averaging.

4.1.2. Pressure and surface tension terms

In the context of pressure-correction algorithm, the GFM also applies to the pressure gradient term to impose the pressure jump condition

$$[p]_I = 2\sigma\kappa \quad (53)$$

as well as the continuity of the velocity field

$$\left[\frac{1}{\rho} \frac{\partial p}{\partial x} \right]_I = \frac{[u^*]_I - [u]_I}{\Delta t} = 0, \quad (54)$$

where u^* and u denote the predicted and corrected (divergence free) velocities, respectively.

Setting $\xi = p$ and $\chi = 1/\rho$, Eq. (51) is used to derive a symmetric Laplacian equation that incorporates the surface tension term [15]. Such a discretization also guarantees a discrete balance of the surface tension forces across the interface, which is known to reduce the amplitude of spurious currents [16]. In the proposed method, the curvature estimation errors are however still present, and the spurious currents are not expected to disappear. Note also that the harmonic averaging of the specific volume $1/\rho$ simply results in a line averaged value of the density field.

In the proposed method, this treatment is slightly modified such that the application of the pressure correction does not introduce any momentum conservation error. This modification and the Ghost Fluid method [15] are however equivalent to second order. The surface tension implementation is identical to the original algorithm.

4.2. Time advancement

The following equations are derived for the backward Euler time advancement scheme. In the staggered system used in this work, the density value over each $\alpha \in \llbracket 1, 3 \rrbracket$ control volume may be updated according to

$$\frac{\rho_\alpha^{n+1} - \rho_\alpha^n}{\Delta t} + \frac{\delta_1 \tilde{\rho}_\alpha^{\epsilon\beta} v_{\epsilon\beta}^\alpha}{\delta_1 x_\beta} = \rho_\alpha^{n+1} \frac{\delta_1 v_{\epsilon\beta}^\alpha}{\delta_1 x_\beta}, \quad (55)$$

where the definitions introduced in Section 3.3.1 are used. Despite the right-hand side in Eq. (55) being of small amplitude and of order Δx^3 [26], in the present method, we recall that it is required for the following prediction-correction scheme to be well conditioned at large density ratios:

$$\begin{cases} \frac{\rho_\alpha^{n+1} u_\alpha^* - \rho_\alpha^n u_\alpha^n}{\Delta t} = \left(\rho_\alpha^{n+1} u_\alpha^* \frac{\delta_1 v_{\epsilon\beta}^\alpha}{\delta_1 x_\beta} - \frac{\delta_1 \tilde{\rho}_\alpha^{\epsilon\beta} u_\alpha^* v_{\epsilon\beta}^\alpha}{\delta_1 x_\beta} \right) + (\text{visc.}) + b_\alpha, \\ \rho_\alpha^{n+1} \frac{u_\alpha^{n+1} - u_\alpha^*}{\Delta t} = - \frac{\delta_1 p}{\delta_1 x_\alpha} + (\text{surf. tens.}). \end{cases} \quad (56)$$

The pressure field p follows from imposing the divergence free condition on (u_α^{n+1}) :

$$\frac{\delta_1 u_\alpha^*}{\delta_1 x_\alpha} = \Delta t \frac{\delta_1}{\delta_1 x_\alpha} \left(\frac{1}{\rho_\alpha^{n+1}} \frac{\delta_1 p}{\delta_1 x_\alpha} - \frac{(\text{surf. tens.})}{\rho_\alpha^{n+1}} \right). \quad (57)$$

In Eq. (56), the viscous effects are accounted for using a harmonic averaging. A noticeable difference in Eq. (57) from the original GFM [15] is the use of ρ_α^{n+1} in place of the line averaged density. This choice leads to a momentum conservative projection step, and, while both formulas are equivalent for linear interfaces, the underlying consistency was found to improve the stability of the flow solver.

The time integration is carried using an explicit second order Runge–Kutta.

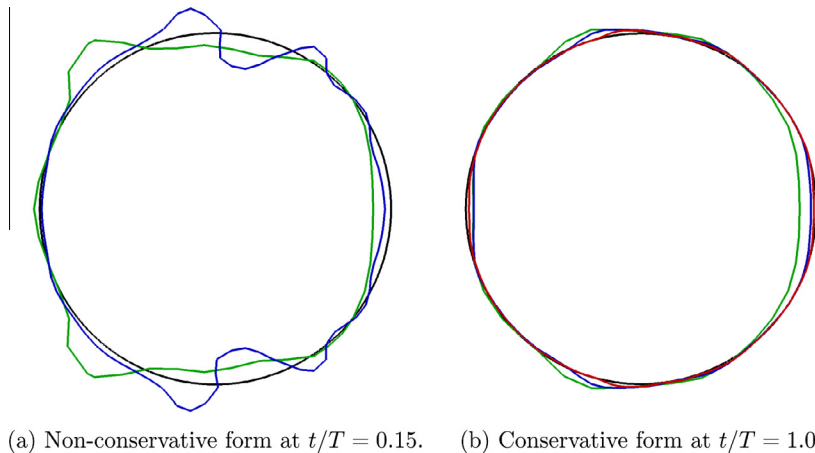


Fig. 5. High density ratio droplet in quiescent gas ($\rho^L/\rho^G = 10^3$): 32×32 mesh (green), 64×64 (blue) and 128×128 (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Verification

5.1. High density ratio drop

This test case was introduced in [19]. It consists in simulating a very large density drop in a low density gas. In the limit of an infinite liquid density, the drop should behave like a hard sphere. It is used here as a way to assess the discretization proposed in this work.

Fig. 5(a) shows how a simple non-conservative formulation (purely centered) quickly leads to unphysical results: after a short time, the infinity norm of the velocity field tends to grow unbounded, therefore leading to a time-step size tending to zero, at which point the computation essentially stalls. The proposed method however remains stable at all time, and the geometric error is seen in Fig. 5(b) to reduce under mesh refinement.

5.2. Standing wave

This test case, proposed by Popinet and Zaleski [31], aims at validating the implementation of the convective, viscous, and capillary transport terms. Based on the analytical solution presented by Prosperetti [32], it consists in comparing the analytical decay of a two-dimensional disturbed interface (initialized here in a $[-\lambda/2, \lambda/2] \times [-3\lambda/2, 3\lambda/2]$ domain as the zero iso-contour of,

$$\varphi(x, y) = y - A_0 \cos\left(\frac{2\pi x}{\lambda}\right), \quad (58)$$

to a numerical solution (A_0 to 0.01λ).

Following [7], the surface tension coefficient is set to $\sigma = 1$ and the density in the heavy fluid to $\rho^l = 1$. Two different cases are studied: the first case uses $\rho^G = 1.0$, and the dynamic viscosity is chosen so as to set the Laplace number

$$\text{La} = \frac{\sigma \rho \lambda}{\mu^2}, \quad (59)$$

to 3000 in both phases. In the second case, which aims at reproducing the physical properties of an air and water mixture, the light fluid density and dynamic viscosity are modified to satisfy $\rho^l/\rho^G = 850$ and $\mu^l/\mu^G = 55.72$ while holding the heavy fluid Laplace number constant (3000).

Given the inviscid natural frequency

$$\omega = \sqrt{\frac{\sigma k^3}{\rho^l + \rho^G}}, \quad (60)$$

where $k = 2\pi/\lambda$, the RMS values of the relative error averaged between $t = 0$ and $25/\omega$ are reported in Tables 1) and 2, and compared to the error levels obtained with the Gerris solver, an open-source adaptive solver based on a non-conservative formulation [33,7]. In both low and high density ratio cases, when the wave is under-resolved, the convergence of the present method is linear, but accelerates as the resolution increases before a saturation of the error appears. Such a saturation was also found in the case of the Level Set method presented in [34]. The error levels are higher than those reported for Gerris, most likely due to the inaccuracies in the curvature computation.

5.3. Sloshing problem

The geometric discretization of the convective transport term introduces an upwind bias in the momentum equation. Despite the centered discretization in the single phase regions of the flow, the numerical dissipation in the interface vicinity is expected to affect the quality of the numerical solution. The following test case is introduced to characterize this precise effect.

A domain of size $[-\lambda/2, \lambda/2] \times [-\lambda, \lambda]$ is used to represent a two-dimensional wave, initialized as the zero iso-level of the function

$$\varphi(x, y) = A_0 \cos\left(\frac{2\pi x}{\lambda}\right) - y, \quad (61)$$

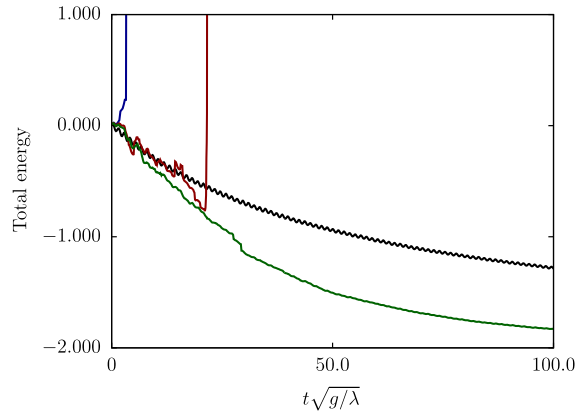
Table 1

Comparison of the RMS error in standing wave case ($\rho^l/\rho^G = 1$, $\mu^l/\mu^G = 1$, $\text{La} = 3000$).

Method/resolution	8	16	32	64	128
Present study	0.355	0.118	0.0274	0.00627	0.00439
Gerris [7]	0.157	0.0279	0.00838	0.0018	0.000545

Table 2Comparison of the RMS error in standing wave case ($\rho^L/\rho^G = 850$, $\mu^L/\mu^G = 55.72$, $La^L = 3000$).

Method/resolution	8	16	32	64	128
Present study	0.304	0.172	0.0461	0.0134	0.0124
Gerris [7]	0.197	0.0754	0.0159	0.00576	0.00313

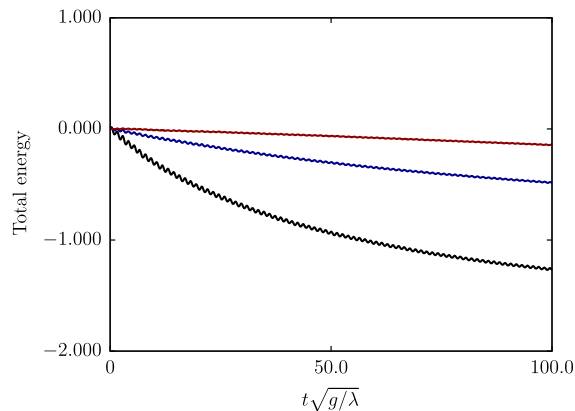
**Fig. 6.** Decay of total energy for the sub-critical sloshing test: comparison of the proposed method (black) and artificial viscosity method ($C = 0.001$, $C = 0.005$, $C = 0.01$).

where $\varphi > 0$ denotes the heavy fluid region. A periodic boundary condition is used along the x -axis. In the y -direction, a negative gravity force is imposed, and a symmetry condition is applied at both ends. Viscosity and surface tension are neglected such that the total energy (potential and kinetic) is analytically conserved. Numerically, the mass error being negligible (less than $10^{-4}\%$ for all cases), the energy decay rate is mainly due to the numerical dissipation of the proposed method.

In the absence of viscous and capillary effects, the state at time $t^* = t\sqrt{g/\lambda}$ is only a function of the density ratio ρ^L/ρ^G (1000 in all cases presented here) and the amplitude A_0 . A first case with small amplitude ($A_0 = 1/8$) is computed, for which the wave does not break. This simple sub-critical configuration, which does not involve any significant topology changes, is used to study the performance of the proposed method with respect to a simple centered non-conservative formulation supplemented by an artificial viscosity method [3] consisting of introducing a high-wavenumber viscosity

$$C\rho\Delta x^{r+1}\left|\frac{\partial^r u}{\partial x^r}\right|. \quad (62)$$

Here, the constant C_μ is tuned until stability of the centered non-conservative computation is reached. The evolution of the total energy is reported for a 128×256 mesh in Fig. 6, where it can be seen that the proposed method introduces signifi-

**Fig. 7.** Decay of total energy for the sub-critical sloshing test: behavior of proposed method under mesh refinement (128×256 , 256×512 , 512×1024).

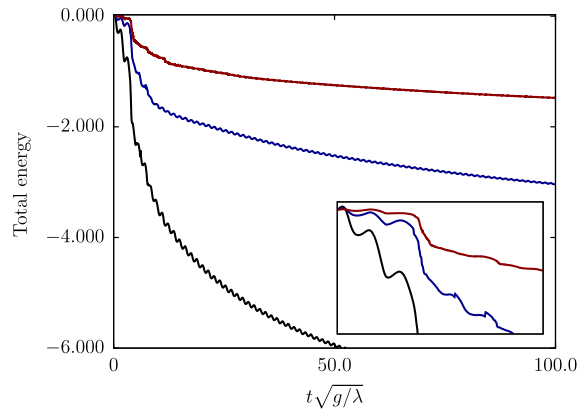


Fig. 8. Decay of total energy for the sup-critical sloshing test: behavior of proposed method under mesh refinement (128×256 , 256×512 , 512×1024).

cantly less dissipation than the optimal artificial viscosity case (obtained for $C = 0.01$, the other computations for $C = 0.005$ and $C = 0.001$ being unstable). The behavior of the proposed method under grid refinement is also examined in Fig. 7, where the decay rate may be seen to be close to second order for this simple sub-critical configuration.

When moving to more complex cases, the numerical dissipation is expected to be closer to a first order effect. This may be seen in Fig. 8, where the amplitude was doubled ($A_0 = 1/4$) in order to introduce more significant topology changes (liga-

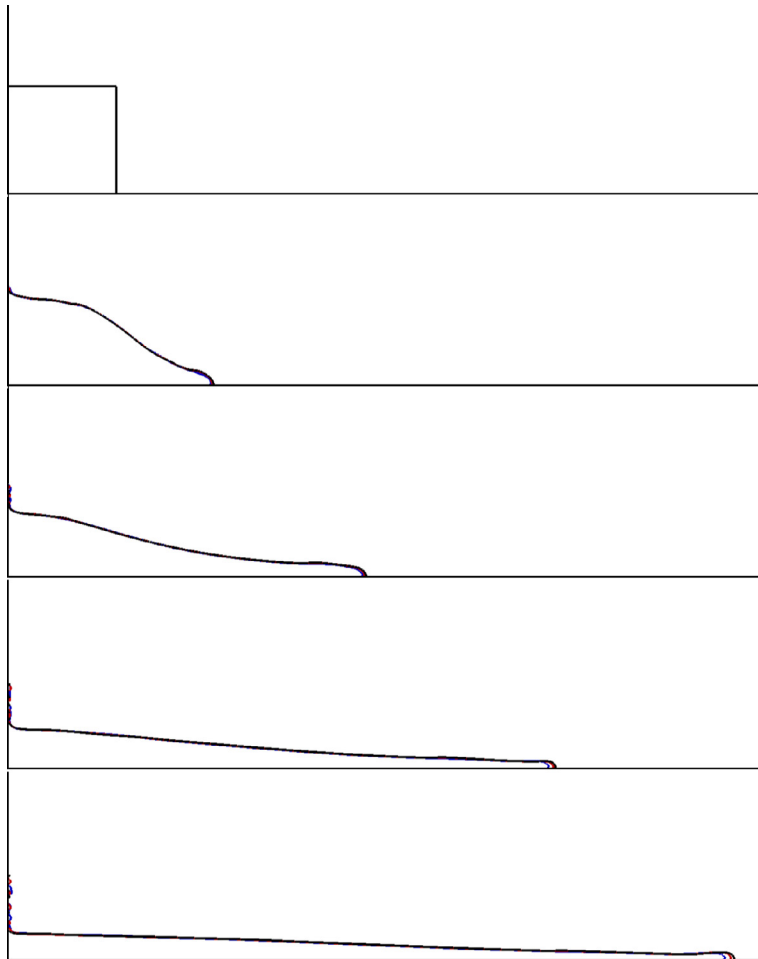


Fig. 9. Interface location at dimensionless times $t\sqrt{g/h}$ of 0.0, 1.0, 2.0, 3.0, and 4.0 (dam breaking): 512×128 mesh (blue), 1024×256 (red), and 2048×512 (black). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

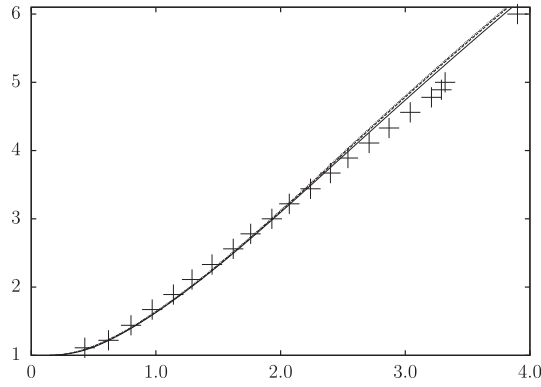


Fig. 10. Distance of surge front from left wall (dam breaking): 512 × 128 mesh (dotted), 1024 × 256 (dashed), and 2048 × 512 (solid).

ment and droplet formation). After the initial transient, a behavior similar to the sub-critical case is seen. Also note that this case turned out to be particularly challenging for the non-conservative formulation with artificial viscosity, which rapidly leads to unphysical wrinkling of the interface and unstable computations.

5.4. Dam breaking

Following [35], this test case is based on the experiments conducted by Martin and Moyce [36], where the collapse of a liquid column onto a flat plate is investigated. In the present computations, the height and width of the column are set to $h = 5.715$ cm, and the domain size is 40 cm × 10 cm. The boundary conditions are no-slip on the left and bottom walls, slip on the top boundary, and convective outflow to the right of the computational domain. The thermodynamic properties are those of air and water ($\rho^L = 1000$ kg/m³, $\rho^G = 1.226$ kg/m³, $\mu^L = 1.137 \times 10^{-3}$ kg/ms, and $\mu^G = 1.78 \times 10^{-5}$ kg/ms), and the gravitational acceleration g is set to 9.81 m/s².

Three different meshes are used, 512 × 128, 1024 × 256, and 2048 × 512, with respective time-steps of 1.5×10^{-4} s, 0.75×10^{-4} s and 0.375×10^{-4} s (chosen for the convective CFL not to exceed 0.5).

As seen from Fig. 9 the agreement between all three meshes is good, especially in the well resolved areas. The mesh resolution is seen to affect mostly the solution in the vicinity of the surge, as also emphasized in Fig. 10. This discrepancy may also be due to a lack of resolution to accurately capture the influence of the wall on the film dynamics.

5.5. Atomization of a water jet in quiescent air

In order to assess the performance of the proposed method in complex interface topologies, the atomization of a round jet of water in quiescent air is computed. The flow at the inlet (diameter D) results from the computation of a scaled-up injector [37] (details provided in [38]). The governing parameters are chosen as follows. The density ratio is set to

$$\frac{\rho^L}{\rho^G} = 800. \quad (63)$$

The Reynolds number, held identical in both phases, is limited to,

$$\text{Re}_D = \frac{\rho \bar{U} D}{\mu} = 5000, \quad (64)$$

where $\bar{U}$ denotes the bulk velocity at the inlet. The liquid Weber number is set to



Fig. 11. Atomization of a water jet in quiescent air: interface location at $t^* = 20$.

$$\text{We}_D^L = \frac{\rho^L \bar{U}^2 D}{\sigma} = 60,000. \quad (65)$$

Lastly, the size of the domain used to simulate the plenum is set to $24D \times 8D \times 8D$. The mesh spacing is chosen so as to guarantee

$$\text{We}_{\Delta x}^G = \frac{\rho^G \bar{U}^2 \Delta x}{\sigma} = 1, \quad (66)$$

hence leading to a mesh of size $1536 \times 512 \times 512$ (uniform spacing), a total of 400 millions cells. The CFL number was set to 0.8.

Fig. 11 shows the interface front at non-dimensional time $t^* = t\bar{U}/D = 20$, reached after 8000 iterations. To reach this instant, the computation, carried on a Dell C6100 Quad-Node system with Intel Westmere-EP processors and Infiniband interconnect, required 135,000 CPU hours shared amongst 1024 processors.

6. Conclusion

A fractional step method was proposed to simulate incompressible free surface flows at arbitrary density ratio. Based on the conservative form of the two-phase Navier–Stokes equations, a geometric discretization of the convective transport is proposed, which guarantees monotonicity preservation for the density, the momentum as well as the velocity fields.

Two algorithms, the forward and the backward time integrations, were presented in the form of two discretely equivalent formulations. The first formulation, based on a Lagrange–Remap interpretation of the convective transport, was used to assess the non-oscillatory properties of both time integrations, and to illustrate the concept of ‘consistency’ between the discretization of the density and momentum transport equations. The second formulation consisted in interpreting the time integrations as geometric upwind Finite Volume discretizations. This viewpoint was then used to identify the origin of the conservation errors, and to formulate a projection equation that is also consistent with the interface transport.

Implemented in a staggered structured Cartesian mesh, the resulting algorithm was shown to remain accurate and stable in different regimes, from capillary dominant flows to high density ratios flows, in the presence of simple as well as complex topologies.

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